

Conflict-Complete Retrieval: Returning Checkable Contradictions from Bounded Agent Memory

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Abstract

An agent cannot act on a contradiction that memory retrieval does not expose. We define *conflict completeness*: for every query-relevant conflict, a retrieval policy returns at least one complete, source-grounded witness that a sound checker can verify from the context alone. We show that every fixed-hop policy based only on IRI adjacency fails this property on a simple literal-keyed construction. We then implement a Conflict Witness Index (CWI) that incrementally maintains key-induced record equivalence and returns one shortest witness per conflict. Controlled evaluation covers 252 instances, including 132 conflicts, with same-record, direct-key, key-chain, hierarchy, and time/scope cases. Exact maintained, exact on-demand, and adaptive exact retrieval return 2–12 witness triples. Direct key-aware retrieval is nearly as compact for non-chain cases, but requires 91–95 triples on two- and three-link chains; BM25, dense, and hybrid retrieval require up to the whole 336–440-triple stores there. On a deterministic 100,000-record workload, CWI uses 1.893 index writes per source triple and processes a 100-query batch in 0.67 ms median over 11 measured repetitions. Detection itself is a standard exact key join, not a contribution. All claims are limited to synthetic data with an oracle schema; the result is a retrieval contract and cost study, not evidence of real-world conflict prevalence.

Introduction

Long-lived agents increasingly move information between a large external memory and a bounded language-model context. Systems such as MemGPT explicitly manage memory tiers (Packer et al. 2023), while graph retrieval systems such as HippoRAG integrate information across stored passages (Gutiérrez et al. 2024). These systems are normally evaluated on whether retrieval supports an answer. We ask a different question: if the store already contains incompatible answers, what must retrieval return so the conflict is not silently hidden?

Consider two records that are declared to identify the same person through an email key, but give different values for a single-valued predicate. A consumer shown only one record cannot distinguish “no conflict exists” from “the

other premise was not retrieved.” Returning a Boolean flag is also insufficient for audit: the consumer cannot check the sources. Returning the whole store is complete but defeats bounded retrieval.

Our contribution is a contract between storage and a downstream reasoner: for each relevant conflict, return one whole minimal witness. Minimal explanations themselves are classical justifications, conflict sets, and why-provenance (Reiter 1987; Kalyanpur et al. 2007; Baader and Peñaloza 2010; Buneman, Khanna, and Tan 2001; Green, Karvounarakis, and Tannen 2007). We do not claim a new conflict detector. The contribution is to make the existence of a checkable witness a completeness property of the *retrieval policy*, prove a boundary for an important policy class, and measure the context and maintenance costs against exact and learned baselines.

The experiments produce three findings. First, a key-aware neighborhood is a strong baseline: for direct conflicts it is usually within a few triples of an exact witness. Second, chained identity is the regime where record-level neighborhood and relevance retrieval become expensive. Third, a maintained index and a fresh exact join return the same witness-sized output; maintenance buys repeated-query latency, not a smaller certificate. These boundaries are central to the claim.

Conflict-Complete Retrieval

Store and schema. An assertion is $a = (s, p, o, \tau, \sigma, \pi)$: an RDF-like triple, a valid-time interval, a scope, and provenance. A confidence threshold θ defines the premise store S_θ . Missing confidence is excluded. Literals use exact term identity; no normalization or entity linker is assumed.

The oracle schema declares a single key predicate k_C for a keyed class C and a set E of predicates that are single-valued per entity, overlapping time, and scope. Records x and y are directly co-keyed, written $x \approx_K y$, when both have type C and the same k_C value. Let $\sim_K$ be its reflexive, symmetric, transitive closure. Bottom time or scope overlaps everything; otherwise time intervals must intersect and scopes must be equal.

A conflict $\gamma = (e, p, \{v_1, v_2\}, \tau, \sigma)$ exists when $p \in E$, $v_1 \neq v_2$, and two records in $[e]_{\sim_K}$ assert the values over overlapping time and scope. This preserve-and-surface semantics differs from repair-based consistent query answering (Are-

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nas, Bertossi, and Chomicki 1999): both assertions remain in the store and the violation becomes an explicit object.

Rule fragment. The executable semantics has three rules. First, the two type and two matching key assertions derive $x \sim y$. Second, equivalence propagates values only for predicates in E ; it is not unrestricted `sameAs` replacement. Third, two unequal E -values derive a conflict exactly when their annotations overlap. The least fixpoint is finite: key equivalence partitions finitely many records, propagation copies only finitely many stored E -values across a partition, and violation generation is pairwise within a partition. The confidence gate filters leaves before any rule fires.

Definition 1 (Witness). *A set $W \subseteq S_\theta$ is a witness for conflict γ if the key, type, value, time, and scope premises in W entail γ , and no proper subset does. Let $\mathcal{W}_\gamma$ be all such witnesses.*

A same-record witness has two value assertions. A direct cross-record witness has two types, two matching keys, and two conflicting values, hence six logical triples. For a shortest chain of m co-key links, intermediate types are shared across adjacent links, giving

$$|W| = (m + 1) + 2m + 2 = 3m + 3. \quad (1)$$

Serialized certificates additionally carry any record-level time/scope metadata required to reproduce the annotated assertions.

Proposition 1 (Detector soundness and completeness). *For the rule fragment above, the least fixpoint derives a conflict if and only if the quotient store under $\sim_K$ contains two unequal values for a predicate in E with overlapping time and scope. Every minimal set of source leaves in such a derivation is a witness.*

Proof. For soundness, key-rule premises satisfy direct co-keying, so propagation moves an E -value only between representatives of the same quotient entity. The violation rule then has exactly the two unequal, overlap-compatible facts required by the conflict semantics. For completeness, take any semantic conflict. Its endpoint records lie in one finite $\sim_K$ class and therefore have a path of direct co-key steps. Repeated propagation carries both endpoint values to one representative, where the violation rule fires. Removing redundant derivation leaves until inclusion-minimal gives a witness. $\square$

A query q has entity seeds $\text{ent}(q)$ and predicates $\text{pred}(q)$. A conflict is relevant if its equivalence class intersects the seeds and its predicate is requested. Write $\Gamma(S_\theta, q)$ for the relevant conflicts.

Definition 2 (Conflict completeness). *A policy $\mathcal{R}$ is conflict-complete for (S, q) iff*

$$\forall \gamma \in \Gamma(S_\theta, q) \exists W \in \mathcal{W}_\gamma : W \subseteq \mathcal{R}(q, S). \quad (2)$$

It is (B, μ) -conflict-complete when Eq. 2 holds and $\mu(\mathcal{R}(q, S)) \leq B$ for a context-cost measure μ .

Equation 2 requires one complete proof per conflict, not every alternative proof. We call the latter, stronger property *proof-exhaustive completeness*. The distinction matters because a conflict may have exponentially many minimal proofs, while one is sufficient for checking. The identity policy $\mathcal{R}(q, S) = S$ is trivially complete; the interesting regime is $B \ll \mu(S)$.

What the contract does and does not guarantee. Conflict completeness is a property of the returned assertions, independent of the checker or language model that consumes them. It prevents retrieval from being the reason a finite, relevant conflict is unverifiable. It does not guarantee that an unsound consumer notices the conflict, that the oracle schema is correct, or that every real inconsistency has a finite witness in this fragment. Conversely, a policy can satisfy ordinary answer relevance while failing Eq. 2; seeing one plausible value is insufficient.

A Boundary for IRI-Only Neighborhoods

Let $G_I(S)$ contain an undirected edge only when two IRIs co-occur as subject and object in a triple. Literals are not graph nodes. A k -hop IRI policy may return assertions incident to IRIs within distance k of a query seed.

Proposition 2 (Fixed-hop incompleteness). *For every finite k , there is a store and query with a relevant six-assertion cross-record witness for which any policy restricted to the k -hop component of $G_I(S)$ is not conflict-complete.*

Proof. Create typed records x and y with the same literal key u , and assert different values of $p \in E$. Seed the query at x . Add no IRI-object path from x to y . Then $x \sim_K y$, so the six assertions form a relevant witness. But y is in a different component of $G_I(S)$ and its type, key, and value assertions cannot be returned at any finite k . $\square$

This result is deliberately narrow. It does not apply to literal-aware, key-aware, lexical, dense, or hybrid policies. On connected benchmark stores, IRI breadth-first search can also become complete by expanding to most or all of the store. The proposition establishes a counterexample class, not a claim about every graph-retrieval workload.

Conflict Witness Index

CWI maintains: (i) buckets indexed by class, key predicate, and exact key value; (ii) union-find classes induced by bucket membership; (iii) single-valued assertions and time/scope annotations per class; and (iv) a record-bucket graph for certificate reconstruction. A touching insertion or metadata update re-evaluates the affected equivalence class. At query time, CWI selects one shortest record-bucket path for each relevant conflict and returns the corresponding source assertions.

Proposition 3 (CWI contract). *Under the oracle schema and exact-literal semantics, after processing a store snapshot, CWI is conflict-complete and each returned certificate is a minimal witness. A conflict whose shortest supporting chain has length m uses $3m + 3$ logical assertions, or two for a same-record conflict.*

Proof sketch. Union-find contains exactly the transitive closure of direct co-key buckets. Re-evaluating every constrained predicate in a touched class enumerates exactly pairs with unequal values and overlapping time/scope. A shortest record-bucket path supplies all and only the type and key premises connecting the endpoint value assertions. Removing an endpoint value breaks the violation; removing a path premise breaks co-reference. Thus the assembled set is minimal and satisfies Eq. 2. $\square$

The proposition is per conflict. When witnesses share assertions, choosing one shortest witness independently need not minimize the size of their global union. CWI is an incremental materialized view (Gupta, Mumick, and Subrahmanian 1993), specialized to retrieval of explanations.

Update and query path. On insertion, CWI records a type, key, constrained value, or annotation. A record enters a key bucket only after both its class and key are known. Joining a non-singleton bucket unions its members and merges their constrained value lists. Every affected predicate cell is then re-evaluated, allowing a later time/scope annotation to retract or restore a conflict in the current record snapshot. Querying a seed locates its union-find root, filters conflicts by the query predicate footprint, and runs breadth-first search on the record-bucket graph only to reconstruct a shortest source path. Output is sorted by stable triple identifier.

Complexity. Let n_c be records in an affected equivalence class and r_p its assertions for constrained predicate p . Bucket insertion has inverse-Ackermann union-find overhead; merging and re-evaluating a touched class is $O(n_c + \sum_p r_p^2)$ in the current pair-enumerating implementation. This quadratic class-local term is a genuine worst case, hidden by the bounded-class generator used below. Index storage is linear in source types, keys, constrained values, and the materialized conflicting pairs. A query is linear in the conflicts returned plus the record-bucket edges visited for their shortest paths. These bounds motivate reporting update amplification and class-controlled scaling rather than claiming store-size-independent maintenance.

Evaluation

Questions. We test: (Q1) which policies return a complete witness and at what context cost; (Q2) whether maintained CWI improves certificate size over exact on-demand retrieval; and (Q3) how its maintenance and lookup costs scale. All fixtures and expected labels are deterministic and synthetic. The schema is supplied as ground truth. This isolates the retrieval mechanism but cannot measure schema induction, extraction errors, or real prevalence.

Fixtures. Eight families contain 252 query instances and 132 conflicts. D20 exercises same-record and direct-key disagreement; XR-small and XR-scale add distracting records; Chain- m requires $m \in \{2, 3\}$ co-key links; H3 varies hierarchy and non-key predicates; and TauSig adds record-snapshot time/scope metadata. Benign controls include shared non-key values, distinct entities, disjoint time, and unequal scope. Chain lengths $m = 0, 1, 2, 3$ are exercised.

Systems. CWI is compared with a fresh full-store exact key/equivalence join per query (*on-demand exact*) and an *adaptive exact* baseline that expands key-aware layers and stops once it can verify a positive certificate. Adaptive search does not generally certify absence at its configured eight-layer cap. Record-level policies retrieve all assertions of selected records: IRI BFS and key-aware neighborhoods at depths 1–4; BM25 (Robertson and Zaragoza 2009); cosine retrieval with `all-MiniLM-L6-v2` at a pinned model revision (Reimers and Gurevych 2019); and BM25–dense reciprocal-rank fusion with $k_{\text{RRF}} = 60$ (Cormack, Clarke, and Buettcher 2009). Relevance methods use $\text{top-}K \in \{1, 2, 4, \dots, 512\}$.

Protocol and implementation. TypeScript systems run under Node.js 22.17.1; the dense baseline uses Sentence Transformers 5.6.1. The encoder weights, exact revision, and Python dependency lock are included in the artifact. Scaling ran on one AMD Ryzen Threadripper PRO 5955WX host. Each fixture, schema, output row, and scale store has a deterministic identifier or SHA-256 digest. Retrieval configurations run over identical serialized stores; the exact checker then evaluates the context returned by each policy. No hosted model call contributes to a confirmatory result. Scale timings use one unrecorded warm-up followed by 11 measured repetitions; we report medians and interquartile ranges (IQR).

We score a conflict only when the retrieved context contains a complete minimal witness. Context cost is serialized triple count; Table 1 reports the smallest configuration that reaches 100% conflict completeness within a family. Values are mean/maximum over conflict queries. Since fixtures and outputs are deterministic, we report exact counts rather than significance tests.

Retrieval Results

Exactness check. Maintained CWI and the full exact equivalence join both produced 132 true positives, 120 true negatives, zero false positives, and zero false negatives. Every emitted witness was rechecked from its returned triples without dereferencing the store. In TauSig, the twelve positive cases include the required metadata premises; 48 controls remain benign because their time or scope cells do not overlap. This is a conformance check for the implementation and fixtures, not evidence that the supplied schema reflects a real domain.

The strongest simple baseline is strong. Key-aware depth one returns every direct conflict in 4–12 triples. Thus, when identity is direct, a maintained witness index saves little context. XR-scale also shows that BM25 can be ideal when the exact key string is present in the query: its eight triples match key-aware retrieval despite a 1,689-triple store. Dense retrieval is not uniformly competitive on exact identifier matching.

Chains expose record-level expansion. For Chain-2 and Chain-3, exact witnesses contain 9 and 12 triples as predicted by Eq. 1. Key-aware retrieval must include the radius ball and averages 94.6 and 91.3 triples. All three relevance methods reach full completeness only at their respective whole-

Family	Instances/conflicts	Store	Exact witness	Key-aware	BM25	Dense	Hybrid
D20	14/8	133	2.0/2	4.0/4	7.5/8	7.8/8	7.4/8
XR-small	16/12	345	6.0/6	8.0/8	83.0/83	345.0/345	163.0/163
XR-scale	64/60	1,689	6.0/6	8.0/8	8.0/8	1689.0/1689	323.0/323
Chain-2	41/10	440	9.0/9	94.6/98	440.0/440	440.0/440	440.0/440
Chain-3	25/6	336	12.0/12	91.3/112	336.0/336	336.0/336	336.0/336
H3-NK1	16/12	405	6.0/6	9.0/9	99.0/99	405.0/405	195.0/195
H3-NK3	16/12	525	6.0/6	11.0/11	244.8/254	525.0/525	493.3/495
TauSig	60/12	648	10.0/10	12.0/12	220.2/226	648.0/648	430.8/433

Table 1: Triples returned at the smallest fully conflict-complete setting (mean/maximum). “Exact witness” is identical for maintained CWI, on-demand exact, and positive adaptive exact output.

Family	CWI	On-demand	Adaptive
D20	.019	.559	.547
XR-small	.012	1.005	.646
XR-scale	.012	2.452	.491
Chain-2	.018	1.302	1.758
Chain-3	.018	1.070	3.408
H3-NK1	.014	1.114	.602
H3-NK3	.015	1.504	.655
TauSig	.026	1.379	.552

Table 2: Median positive-query time in ms, single run. CWI excludes ingest; other methods maintain no state. All return the same witness triples.

store contexts (440 and 336 triples). This does not show that learned retrieval cannot be trained for the task; it shows that standard relevance ranking provides no completeness contract on these controlled chains.

Maintenance does not shrink the certificate. CWI, on-demand exact, and adaptive exact return the same shortest witness on every conflict. Therefore CWI has no output-size advantage over fresh exact computation. Adaptive expansion also finds witness-sized positive answers without maintained state, though it scans increasing layers and its finite cap does not prove benign absence. The operational choice is between write-time maintenance, query-time exact work, and absence guarantees.

Query-time tradeoff. Table 2 separates output equality from computation. It reports median positive-query wall time from the same uncontrolled run. CWI has already paid ingestion; on-demand exact repeats a full join; adaptive exact expands until it finds a proof. The numbers explain the purpose of maintenance, but are not a fair end-to-end throughput comparison because the systems pay setup at different phases.

External Topology Audit

To test whether key chains are solely a generator artifact, we separately audit the CC-BY release of the established DBLP–Google Scholar entity-resolution benchmark (Papadakis 2022). Its 5,347 gold match edges connect 7,626 records into 2,351 components (maximum size 37). Closing those gold edges transitively, 138 components contain 350

record pairs whose two nonempty publication-year strings are unequal. Shortest gold-path depths are 1 for 157 pairs, 2 for 155, 3 for 23, and 4 for 15. Thus 193 disagreement pairs are connected only through a chain, corresponding to 6-, 9-, 12-, and 15-triple logical witnesses under our encoding.

This is an exploratory topology check, outside the 252 confirmatory instances. An unequal year is not an independently adjudicated semantic conflict: publication versions, linkage mistakes, or source metadata errors may explain it. The audit supports only the narrow statement that transitive co-reference and cross-source disagreement co-occur in an established record-linkage benchmark. It does not estimate conflict prevalence in agent memory. The artifact records source hashes and a built-in-only reproduction script.

Scaling Ledger

We generate stores up to 100,000 persons (560,009 triples) and issue 100 queries sampled from 50 persons at each size. Table 3 reports a repeated-run companion to the registered deterministic ledger. The exact join includes its file-read/setup path, whereas CWI ingest starts from parsed triples; their columns are intentionally not presented as an end-to-end speedup.

Index writes remain 1.893–1.895 per source triple; at 100,000 persons the index contains 440,006 entries. The 100-query batch remains below 0.75 ms median at every size. These results support bounded update amplification and fast lookup on this generator. They do not establish production latency, concurrency behavior, memory residency, or a break-even query rate.

Related Work

Justifications and axiom pinpointing identify minimal premises for an entailment (Kalyanpur et al. 2007; Baader and Peñaloza 2010); diagnosis uses minimal conflict sets (Reiter 1987); and database provenance tracks which source tuples support an output (Buneman, Khanna, and Tan 2001; Green, Karvounarakis, and Tannen 2007). CWI reuses these ideas and incremental view maintenance (Gupta, Mumick, and Subrahmanian 1993). Its novelty is not the witness object or exact join, but Eq. 2 as a bounded retrieval contract and the comparative cost analysis.

Long-term LLM memory work studies tiered context management and multi-hop retrieval (Packer et al. 2023; Gutiér-

Persons	Triples	Writes/triple	CWI ingest ms	100 queries ms	Exact join-X ms
300	1,689	1.895	1.20 [0.89, 1.39]	0.75 [0.53, 1.14]	3.08 [2.71, 3.92]
1,000	5,609	1.894	4.02 [3.36, 4.57]	0.49 [0.46, 0.64]	6.28 [6.16, 6.44]
3,000	16,809	1.893	7.92 [7.69, 8.66]	0.39 [0.37, 0.42]	26.40 [25.56, 27.89]
10,000	56,009	1.893	40.59 [39.23, 44.97]	0.55 [0.52, 0.61]	104.15 [101.19, 115.74]
100,000	560,009	1.893	672.20 [644.72, 838.01]	0.67 [0.66, 0.69]	1590.24 [1576.23, 1618.28]

Table 3: Scale results: median [IQR] over 11 measured runs after one warm-up. The query cell is a batch of 100 maintained-index lookups. Exact join-X is a fresh transitive key-equivalence join. No run had a missing or spurious conflict.

rez et al. 2024). Sparse and dense retrieval optimize relevance rather than worst-case witness inclusion (Robertson and Zaragoza 2009; Reimers and Gurevych 2019). Reciprocal rank fusion combines rankings without creating a logical completeness guarantee (Cormack, Clarke, and Buettcher 2009). These objectives are complementary: relevance can select candidate entities, after which conflict-complete retrieval can attach checkable exception evidence.

Limitations and Validity

The strongest limitation is external validity. Every confirmatory result uses synthetic fixtures and an oracle schema. A post-audit of exploratory public data found that an earlier vulnerability analysis had incorrectly treated a non-transitive “related” field as identity, while a clinical-trial probe had treated shared grant identifiers as study identity. We removed both prevalence claims rather than substitute a weaker number. The corrected vulnerability alias graph contained no key chains and lacked semantic conflict labels. The DBLP-Scholar audit establishes chained gold-match topology but does not validate its year disagreements as semantic conflicts. We therefore make no claim that the chain-conflict regime is common.

The formal fragment has one exact key per class, exact literal comparison, single-valued constraints, interval overlap, and equal-or-unspecified scope. It excludes learned entity resolution, composite keys, uncertainty propagation, n-ary constraints, and open-world conflicts without finite witnesses. Confidence and schema are trusted inputs. Record-snapshot time/scope metadata is implemented, but arbitrary event histories are not.

The index is insertion-oriented; deletion, persistence, concurrent updates, and crash recovery remain engineering work. Per-conflict shortest witnesses need not minimize their union. The learned baselines use one frozen encoder without task-specific training. Finally, a complete witness guarantees that a sound checker *can* verify a conflict, not that a language model will notice it. Our exploratory reader runs confounded witness size with distractors and used an outcome-defined competence subgroup, so we exclude them from confirmatory evidence.

Conclusion

Conflict completeness makes a missing contradiction a falsifiable retrieval failure: every relevant conflict must arrive with one checkable witness. Fixed-hop IRI neighborhoods cannot guarantee it, while exact maintained and on-demand

methods can return witness-sized contexts. Simple key-aware retrieval is already competitive for direct conflicts; maintenance is most useful for repeated queries over chained identity or when absence must be certified. Establishing whether those regimes occur in real agent memories, under independently validated schemas and conflict labels, is the necessary next step.

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